

school_statistics

school_infrastructure

social_category_enrollment

SQL PROJECT DOCUMENTATION

School Infrastructure & Enrollment Analysis

A structured SQL analytics project exploring school infrastructure, enrollment trends, and social category distribution across Indian states using joins, CTEs, window functions, and aggregate analysis.

30 Business Questions Solved

3 Linked Datasets

Basic → Intermediate → Advanced SQL

Project Overview

This project analyzes school education statistics using three datasets: **school_statistics**, **school_infrastructure**, and **social_category_enrollment**. The objective is to answer real-world business questions using core and advanced SQL concepts including filtering, aggregation, joins, CASE expressions, CTEs, subqueries, and window functions.

- Filtering
- Aggregation
- Joins
- CASE Expressions
- CTEs
- Subqueries
- Window Functions

Dataset & Column Meaning

school_statistics

Column	Description
State	State/UT name.
Total number of Schools	Total schools in the state.
Enrolments	Total students enrolled.
Average Enrolments Per School	Average students per school.
Pupil Teacher Ratio	Average students per teacher.
Schools with Single Teachers	Number of single-teacher schools.
Schools with Zero Enrolments	Schools having no students.

school_infrastructure

Column	Description
Internet Facility	Schools with internet access.
Computer Facility	Schools having computers.
Functional Computer Facility	Schools where computers are functional.
Electricity / Functional Electricity	Electricity availability and working condition.
Drinking Water / Functional Drinking Water	Water availability and functionality.
Girls Toilet / Functional Girls Toilet	Availability and working condition.
Ramp for CWSN	Accessibility ramps for children with special needs.

social_category_enrollment

Column	Description
General, OBC, SC, ST, Muslim	Student enrolment by social category.

Questions Solved

30 business questions were answered across three progressive difficulty tiers, moving from basic filtering to advanced window-function analytics.

● Basic SQL 1. Top 10 states by schools 2. States with PTR > 30 3. States with zero-enrolment schools 4. Top 5 states with internet facility 5. Andhra Pradesh social category enrolment 6. National average PTR 7. Average enrolment per school 8. Total states/UTs 9. States with fewer than 1,000 schools 10. Top 5 states by enrolment	● Intermediate SQL 1. Highest % single-teacher schools 2. Lowest internet coverage 3. States above average OBC share 4. Girls' toilet maintenance gap 5. PTR bands using CASE 6. Count of PTR bands 7. Highest non-functional computer facility 8. SC share above 20% with PTR 9. States with >90% electricity & drinking water 10. Rank by average enrolment per school	● Advanced SQL 1. Composite infrastructure score 2. Rank states by school count 3. Estimated ST enrolment 4. CWSN ramp quartiles 5. LAG difference in school count 6. Muslim share vs internet 7. Worst computer facility gap 8. PTR vs national average 9. Composite opportunity score 10. Percentile rank of internet coverage
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SQL Concepts Demonstrated

SELECT	WHERE	ORDER BY	GROUP BY	HAVING	Aggregate Functions	JOIN	
CASE	Subqueries	CTEs	RANK	LAG	NTILE	PERCENT_RANK	AVG OVER

Key Insights

1

States differ significantly in school infrastructure and enrolment.

2

Higher internet coverage generally indicates better digital readiness.

3

High PTR values may indicate teacher shortages.

4

Single-teacher schools can affect education quality.

5

Zero-enrolment schools may need policy intervention.

6

Functional infrastructure is more meaningful than total infrastructure.

7

Social category analysis highlights enrolment distribution.

8

Window functions help compare state performance.

9

CTEs simplify complex calculations such as composite scores.

10

The project demonstrates practical SQL skills used in analytics.

Tech Stack Summary

Layer	Details
Language	SQL (Standard / PostgreSQL-compatible syntax)
Datasets	school_statistics, school_infrastructure, social_category_enrollment
Core Techniques	Filtering, Aggregation, Joins, CASE logic
Advanced Techniques	CTEs, Subqueries, Window Functions (RANK, LAG, NTILE, PERCENT_RANK)
Question Tiers	10 Basic • 10 Intermediate • 10 Advanced